Universidad del País Vasco Euskal Herriko Unibertsitatea

COURSE GUIDE 2026/27

Faculty 226 - Faculty of Informatics **Cycle** .

Degree GINFOR20 - Bachelor's Degree in Informatics Engineering **Year** Fourth year

COURSE

26256 - Distributed Systems **Credits, ECTS:** 6

COURSE DESCRIPTION

"Distributed Systems" is an elective subject included in the curriculum of the Degree in Informatics Engineering and is taught (in English) in the fourth year.

The development of information technology has led to the interconnection of individual computers into networks that allow mutual access to shared resources. The next step is the integration of resources and services in distributed systems, which provide the user and the applications with interoperability and transparent access. This course focuses on the concepts used in distributed systems and the mechanisms that allow to support the integration of services and resources.

COMPETENCIES/LEARNING RESULTS FOR THE SUBJECT

At the end of the course, the students should be able to:
- Know the characteristics and internal structure of distributed systems
- Design, implement and evaluate the performance of distributed algorithms and systems

Other important aspects that are going to be promoted are teamwork, the ability to find the information and tools necessary to solve the problems that arise, and the ability to accurately describe the functionality of the developed solutions, so that they can be used by other users or developers.

Theoretical and Practical Contents

Topic 1. Introduction to distributed systems

Motivation. Properties of distributed systems. Distributed applications. Mobile and ubiquitous computing. Hardware support. Operating system support. Communication support. Open systems. Structure of a distributed system. Practice: remote procedure calls, remote method invocation, one-to-many communication with IP Multicast.

Topic 2. Time, causality and global state

Physical time. Clock synchronization. Logical time and ordering of events. Global state and consistency. Practice: clock synchronization tools and protocols (e.g., NTP), synchronization algorithms.

Topic 3. Distributed coordination

Distributed mutual exclusion. Election algorithms. Group communication. Replication. Distributed transactions. The Consensus problem. Practice: group communication with middleware (JGroups, Spread...).

Topic 4. Distributed file systems

Properties of distributed file systems. System model. Name service. File service. High performance file systems. Internet file systems. Examples: NFS, AFS, Coda, CIFS. Practice: test of distributed file systems, and evaluation of their behavior in different situations.

Topic 5. Security in distributed systems

Threats and attacks. Security policies. Security mechanisms. Authentication protocols. Examples: Needham-Schroeder, Kerberos. Secure Multiparty Computation. Practice: test of Kerberos.

Advanced topics: Fault-tolerant systems. Blockchain. Distributed shared memory. Cluster computing. Peer-to-peer networks. High availability systems. Grid computing. Mobile and ubiquitous computing. Practice: development of a group work in the field of distributed systems.

TEACHING METHODS

Although the course is based on the theoretical concepts that support distributed systems, it will have a large practical component. For this, various teaching methodologies will be used, from master classes to more active methodologies such as PBL (Project Based Learning) or pBL (Problem Based Learning) and laboratory activities.

TYPES OF TEACHING

Types of teaching	M	S	GA	GL	GO	GCL	TA	TI	GCA
Hours of face-to-face teaching	40			20					
Horas de Actividad No Presencial del Alumno/a	60			30					

Legend: M: Lecture-based | S: Seminar | GA: Applied classroom-based groups | GL: Applied laboratory-based groups | GO: Applied computer-based groups | GCL: Applied clinical-based groups | TA: Workshop | TI: Industrial workshop | GCA: Applied fieldwork groups

Evaluation methods

- Continuous evaluation
- End-of-course evaluation

Evaluation tools and percentages of final mark

- The percentages and types of evaluation are specified in the following sections. 100%

ORDINARY EXAMINATION PERIOD: GUIDELINES AND OPTING OUT

The exams and assessment activities for this course will be conducted in accordance with the principles of academic integrity, honesty, and ethical conduct established in EHU regulations. Upon enrolling at EHU, students signed the Declaration of Commitment to Ethical Behavior and Academic Integrity pledging not to engage in plagiarism, academic fraud, or use unauthorized materials or devices -including artificial intelligence tools when not expressly permitted- in exams and academic assignments. Failure to comply with these obligations may result in the academic and disciplinary consequences provided for in current university regulations.

CONTINUOUS ASSESSMENT

The evaluation systems that are contemplated are the continuous evaluation system and the final evaluation system. The continuous assessment system is the one that will be used preferably, as indicated in the current regulations of the UPV/EHU. The student who, fulfilling the conditions to continue in the continuous evaluation system, decides to opt for the final evaluation, must inform the teacher within the terms and manner indicated below: via eGela, through the corresponding link, at the latest before the 10th week of the teaching period.

The continuous evaluation mode is based on the following three evaluation tests:
- Individual evaluation questionnaires: 48%
- Practical Works, in groups of three students: reports of the different activities, developed code, specifications, verification results of the practice proposals and interview: 32%
- Project, in groups of three students: 20% (out of which 2% is individual)

To summarize, 50% of the grade is based on individual work and 50% on group work.

To pass the subject it is necessary that the student:
- Completes and submits all assessment tests
- Obtains at least 40% of the evaluation of each of the tests
- The final average mark with all the evaluation tests is at least 5 out of 10

FINAL EVALUATION

For those who do not follow the continuous evaluation, the following evaluation mechanism is foreseen:
- Final individual written test (theoretical questions, practical exercises, analysis/design of algorithms...): 80%
- The realization and delivery of a practical work and an interview about it: 20%

To pass the subject it is necessary that each student:
- Completes and submits all assessment tests
- Obtains at least 40% of the evaluation of each of the tests
- The final average grade with all the tests is at least 5 out of 10

EXTRAORDINARY EXAMINATION PERIOD: GUIDELINES AND OPTING OUT

In the extraordinary call evaluation will be similar to the final evaluation of the ordinary call:
- Final individual written test (theoretical questions, practical exercises, analysis/design of algorithms...): 80%
- The realization and delivery of a practical work and an interview about it: 20%

To pass the subject it is necessary that each student:
- Completes and submits all assessment tests
- Obtains at least 40% of the evaluation of each of the tests

- The final average grade with all the tests is at least 5 out of 10

MANDATORY MATERIALS

The documentation in eGela: https://egela.ehu.eus/

BIBLIOGRAPHY

Basic bibliography

G. Coulouris, J. Dollimore, T. Kindberg, G. Blair. Distributed Systems: Concepts and Design (5th edition). Pearson, 2012.

A. Tanenbaum, M. Van Steen. Distributed Systems: Principles and Paradigms (2nd edition). Pearson, 2007.

Detailed bibliography

S. Mullender. Distributed Systems (2nd edition). Addison-Wesley, 1993.

A. Tanenbaum. Distributed Operating Systems. Prentice-Hall, 1995.

P. Verissimo, L. Rodrigues. Distributed Systems for System Architects. Kluwer Academic Publishers, 2001.

H. Attiya, J. Welch. Distributed Computing: Fundamentals, Simulations, and Advanced Topics. Wiley-Interscience, 2004.

K. Birman. Reliable Distributed Systems: Technologies, Web Services, and Applications. Springer, 2005.

G. Tel. Introduction to Distributed Algorithms (2nd edition). Cambridge University Press, 2000.

N. Lynch. Distributed Algorithms. Morgan Kaufmann Publishers, 1996.

C. Cachin, R. Guerraoui, L. Rodrigues. Introduction to Reliable and Secure Distributed Programming. Springer, 2011.

N. Santoro. Design and Analysis of Distributed Algorithms. Wiley, 2007.

Journals

IEEE Transactions on Parallel and Distributed Systems

(Springer) Distributed Computing

(Elsevier) Journal of Parallel and Distributed Computing

ACM Transactions on Computer Systems

IEEE Transactions on Computers

Web sites of interest

www.cdk5.net

www.distributed-systems.net

http://www.ntp.org/

http://www.jgroups.org/

http://www.spread.org/

http://web.mit.edu/Kerberos/

OBSERVATIONS

AVISO: ESTA ASIGNATURA SE OFERTA SOLAMENTE EN INGLES.